



UNIVERSITY OF COLOMBO, SRI LANKA



UNIVERSITY OF COLOMBO SCHOOL OF COMPUTING

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

Second Year Examination - Semester II - UCSC AY21 [2025] (Held in March/April 2025)

SCS 2212 – Automata Theory

TWO (2) HOURS

Answer ALL questions

To be completed by the candidate

Examination Index No:

Important Instructions to candidates:

- I. Students should answer in the medium of English language only using the space provided in this question paper
- II. Note that questions appear on both sides of the paper. If a page or a part of this question paper is not printed, please inform the supervisor immediately.
- III. Write your index number CLEARLY on each and every page of the Question paper.
- IV. This paper consists of 4 questions on 12 pages (including the Cover Page).
- V. Answer ALL questions.
- VI. Calculators and any electronic device capable of storing and retrieving text including electronic dictionaries, smart watches and mobile phones are NOT ALLOWED.
- VII. Do not tear off any part of this answer book. Under no circumstances may this book, used or unused, be removed from the Examination Hall by a candidate.

To be completed by the examiners

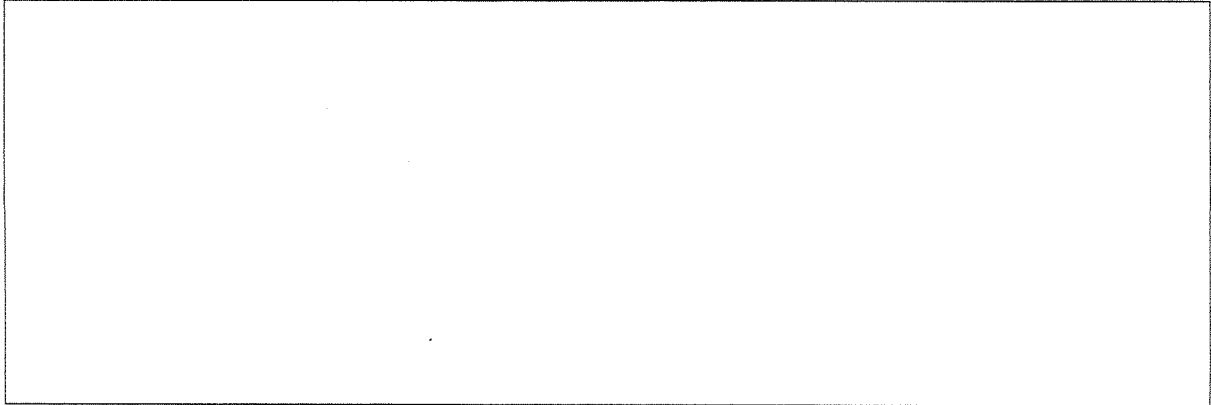
Question No	Marks
1	
2	
3	
4	
Total	

Index Number:

Question 1

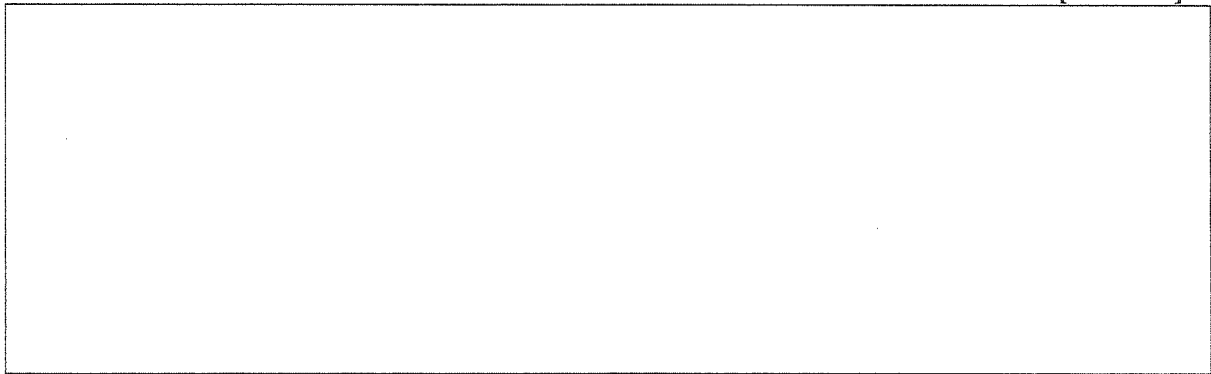
- (i) An electric bulb can be in either in an on state or in an off state. Draw a transition graph to represent the states and transitions of an electric bulb based on the state of switch.

[5 marks]



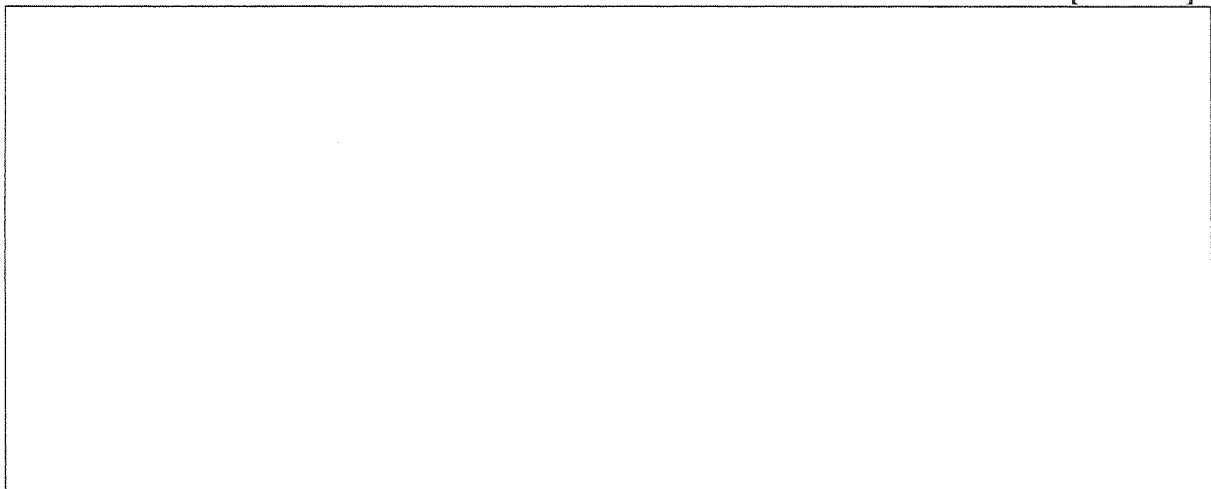
- (ii) State three (3) differences between deterministic finite automata and non-deterministic finite automata.

[5 marks]



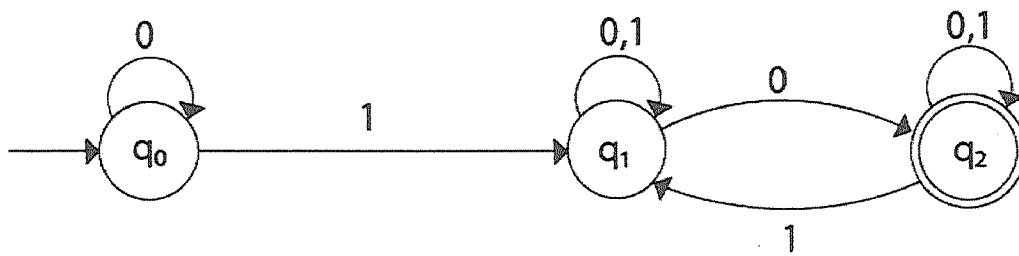
- (iii) State two (2) conditions that are to be satisfied for two finite-state automata to be equivalent.

[5 marks]



Index Number:

- (iv) Convert the following finite automaton to a deterministic finite automaton.



[10 marks]

Index Number:

Question 2

- (i) Express formally, the relationship between finite automata and regular expressions.

[5 marks]

- (ii) Write a regular expression for all binary number strings that ends with 00.

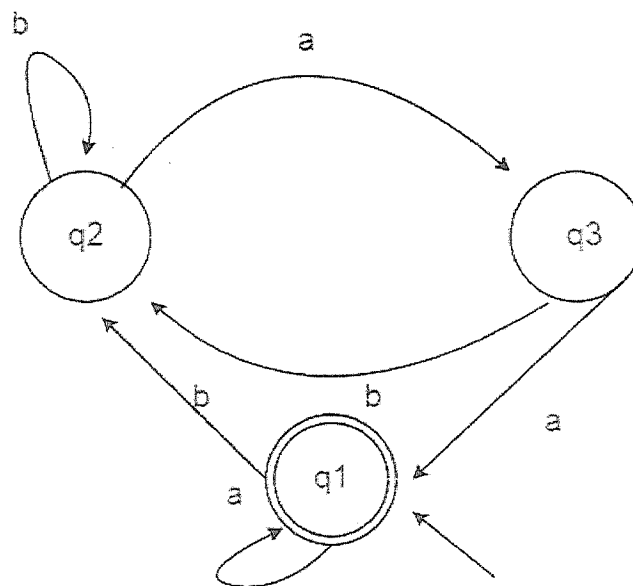
[5 marks]

- (iii) If L_1 and L_2 are regular languages over the same alphabet Σ , can $L_1 \cup L_2$ be considered a regular language over Σ ? Justify your answer.

[7 marks]

Index Number:

- (iv) State Arden's Theorem for generating a regular expression from finite automata. By applying Arden's theorem, generate a regular expression for the following automaton.



[8 marks]

Index Number:

Question 3

- (i) Consider the context-free grammar with productions

$$S \rightarrow AB$$

$$A \rightarrow aA \mid \lambda$$

$$B \rightarrow ab \mid bB \mid \lambda$$

Every derivation of a string in this grammar must begin with the production $S \rightarrow AB$. Clearly, any string derivable from A has only one derivation from A, and likewise for B. Therefore, the grammar is unambiguous. Is this statement TRUE or FALSE? Explain. [5 marks]

- (ii) Consider the grammar with productions

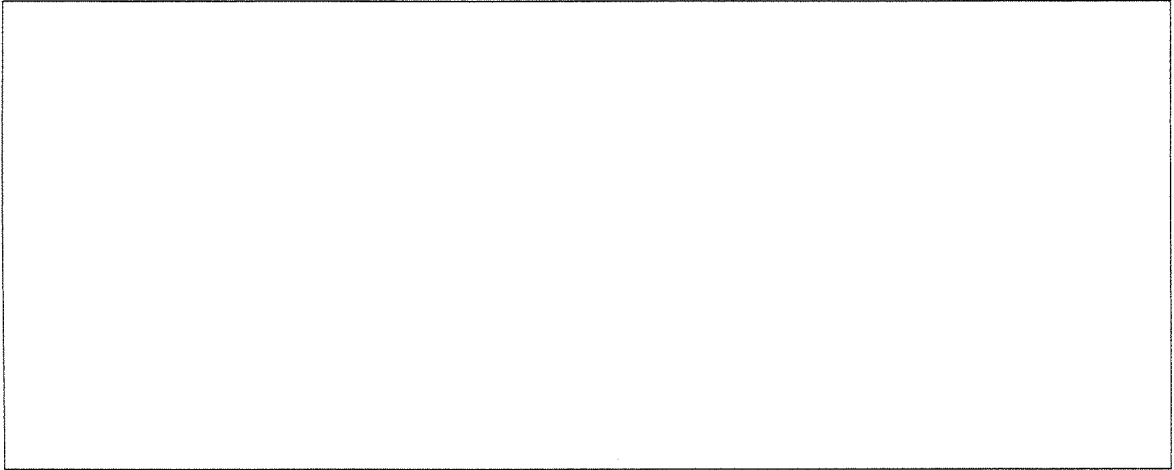
$$S \rightarrow aaB$$

$$A \rightarrow bBb \mid \lambda$$

$$B \rightarrow Aa.$$

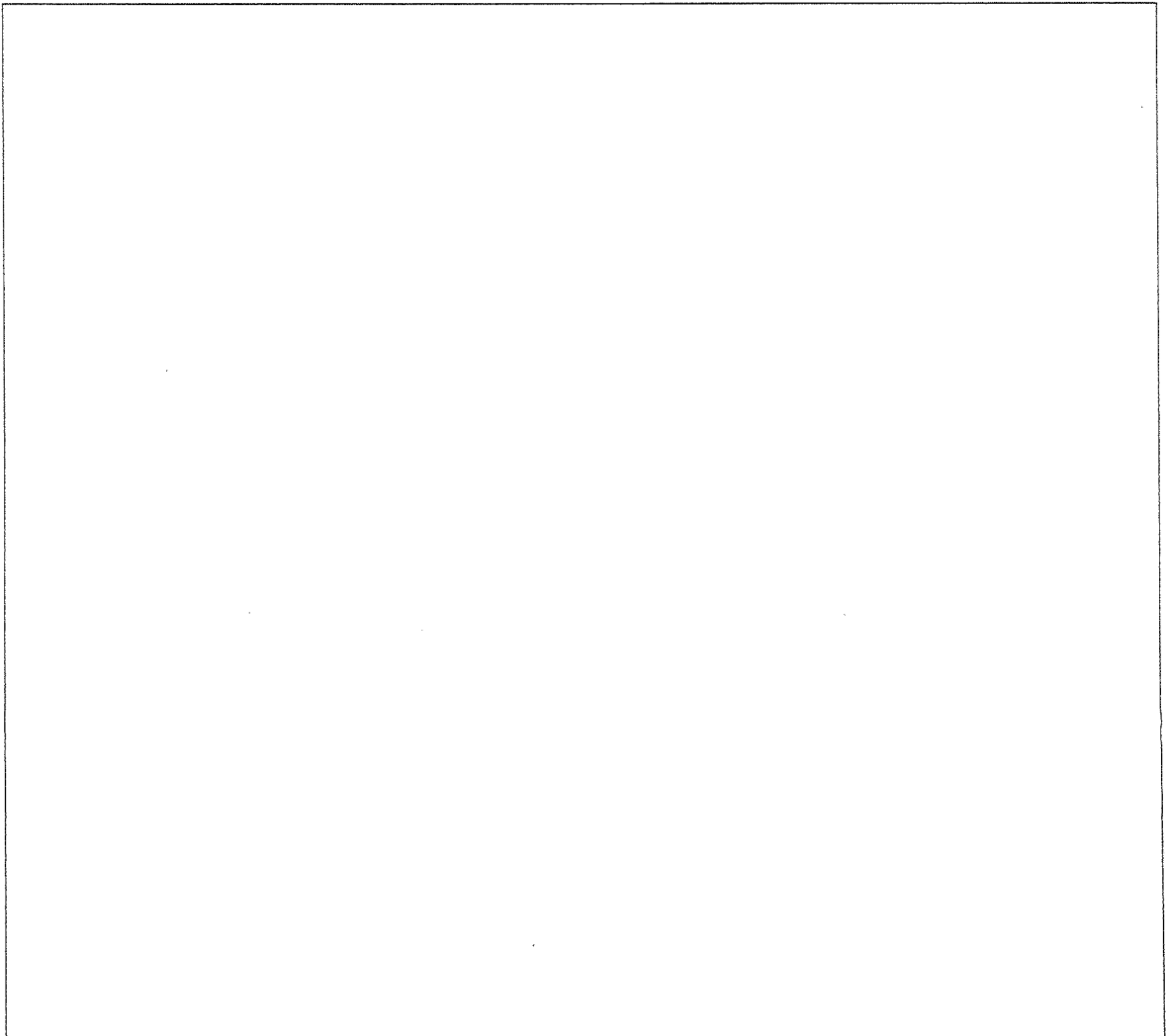
Show that the string **aabbabba** is not in the language generated by this grammar. Identify the **language accepted** by this grammar. [5 marks]

Index Number:



- (iii) Let $\Sigma = \{ (,), [,] \}$, $L = \{\text{properly nested strings from } \Sigma^* \text{ /balanced parenthesis}\}$. For example: $[] \dots [[]]$, $() \dots (())$, $([])$, $[()]$, $[] ()$ are in L , but strings such as $[(])$, $([])$ are not. Construct a Context Free Grammar to accept the language L .

[7 marks]



Index Number:

(iv) Transform the following grammar with productions into Chomsky normal form.

$S \rightarrow baAB$

$A \rightarrow bAB \mid \lambda$

$B \rightarrow BAa \mid A \mid \lambda$

[8 marks]

Question 4

- (i) Explain why Finite Automata cannot recognize all context-free languages. Provide an example to support your explanation. **[3 Marks]**

- (ii) Formally express the language accepted by the following Pushdown Automata (PDA).

$M = (\{Q_0, Q_1, Q_2, Q_3, Q_4, Q_5\}, \{a, b\}, \{0, 1, z\}, \delta, Q_0, z, \{Q_5\})$, with

$\delta(Q_0, b, z) = \{(Q_1, 1z)\}$, $\delta(Q_1, b, 1) = \{(Q_1, 11)\}$, $\delta(Q_1, a, 1) = \{(Q_2, \lambda)\}$,
 $\delta(Q_2, a, 1) = \{(Q_2, \lambda)\}$, $\delta(Q_2, a, z) = \{(Q_3, z)\}$, $\delta(Q_3, a, z) = \{(Q_4, z)\}$,
 $\delta(Q_4, \lambda, z) = \{(Q_5, z)\}$

(Hint: Prove the correctness of your answer by drawing a PDA)

[5 Marks]

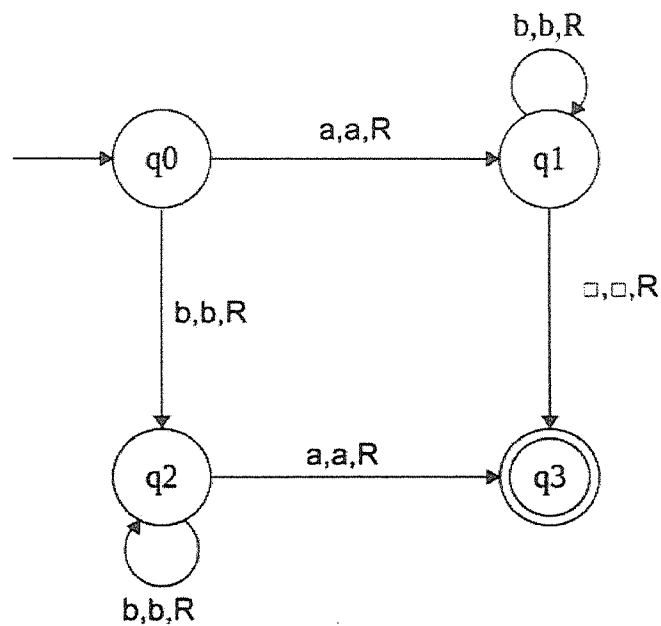
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- (iii) Construct a non-deterministic PDA that accepts the following language on $\Sigma = \{a, b, c\}$:

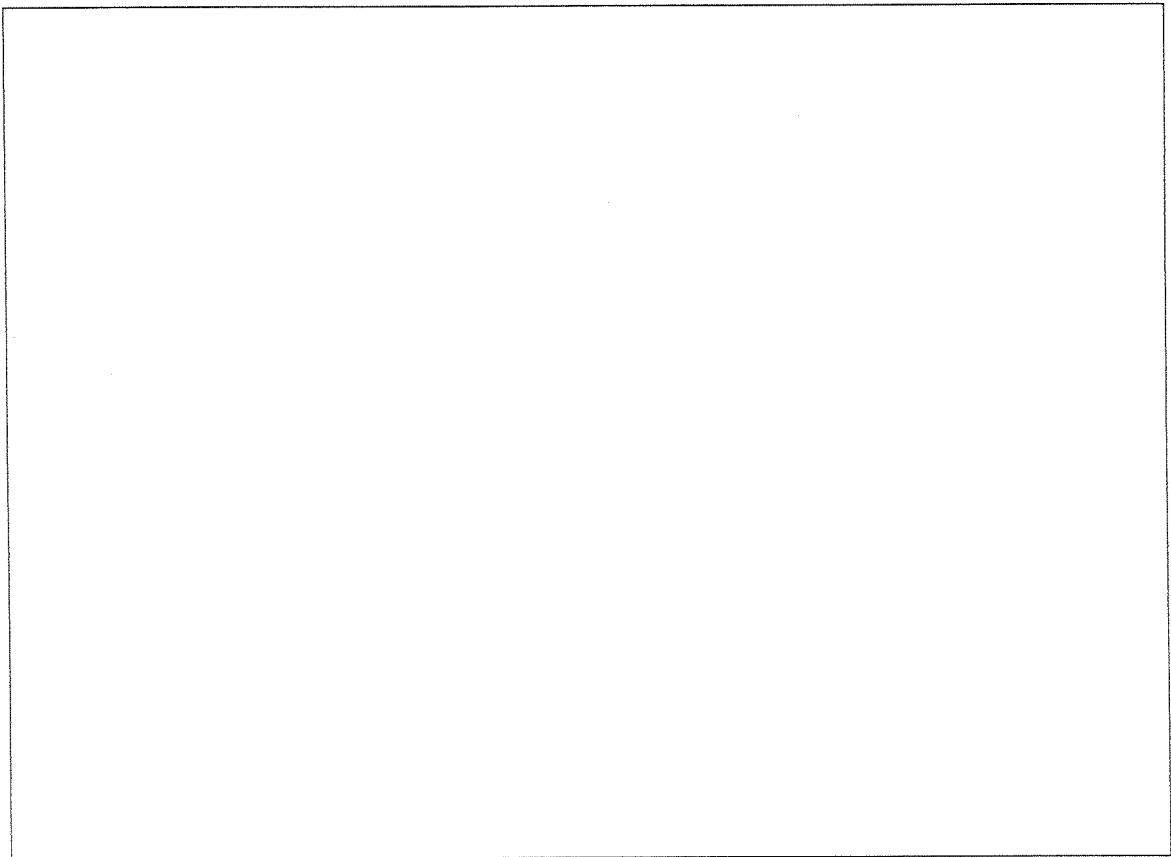
$$L = \{w : n_a(w) + n_b(w) = n_c(w)\}.$$

where $n_a(w)$, $n_b(w)$, $n_c(w)$ are number of a's, b's and c's in string w respectively. [7 Marks]

- (iv) Express formally, the language accepted by the Turing Machine whose transition graph is in the figure below. [3 Marks]



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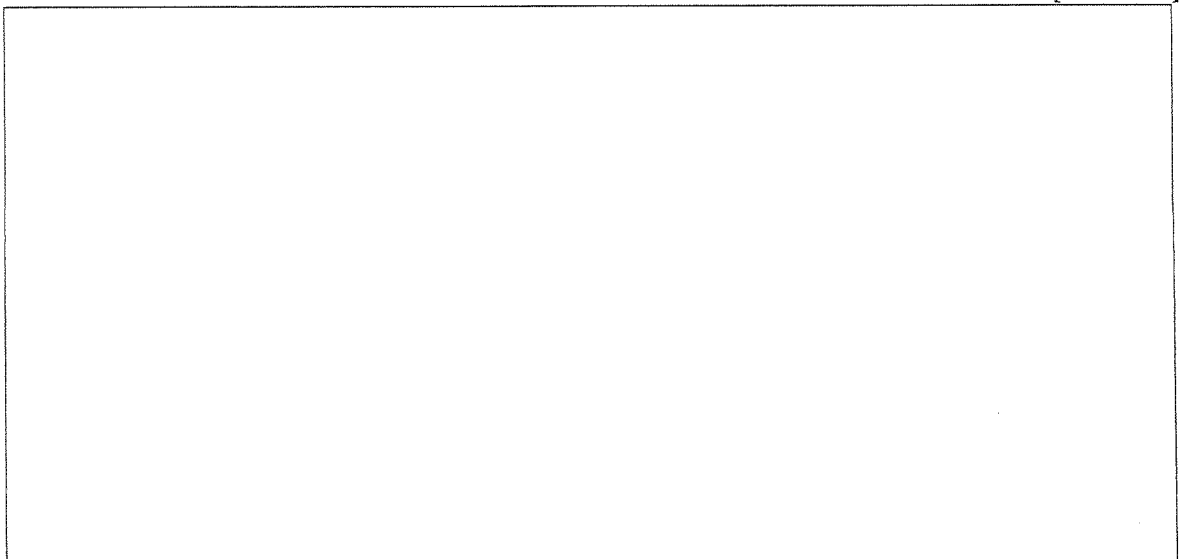


- (v) Construct a Turing machine that will accept the following language on $\{a, b\}$:

$$L = \{a^n b^{2n} : n \geq 1\}$$

(Hint: Draw a transition graph; then check your answer by tracing several test examples)

[7 Marks]



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